MILITARY SPECIFICATION

ASSEMBLY, ANTENNA BOX

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Antenna Box Assembly used on the Atomic Demolition Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder, Tin Alloy, Lead Tin Alloy and Lead Alloy

MILITARY

MIL-S-6872 - Soldering Process, General Specification for
MIL-T-152 - Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical
MIL-V-173 - Varnish, Moisture and Fungus Resistant (for The Treatment of Communication, Electronic and Electrical)

STANDARDS

MILITARY

AN-505-5 - Screw
AN-505-6 - Screw
AN-507-440-12 - Screw
AN-507-632-4 - Screw
AN590C2 - Washer
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Number of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components
MS35233-30 - Screw
MS35430-1 - Terminal, Lug

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MS35430-4 - Terminal, Lug
MS35430-10 - Terminal, Lug
MS35649-22 - Nut
MS35649-62 - Nut

DRAWINGS

ORDNANCE CORPS
8851572 - Assembly Antenna Box
8851573 - Cover Assembly
8851574 - Cover
8851577 - Plate, Connector
8851578 - Receptacle
8851580 - Feed Thru Insulator
8851584 - Assembly, Coil
8851585 - Coil Block
8851586 - Form Var Wire
8851744 - Spacer
8851747 - Variable Capacitor
8851773 - Shielding Tinned Copper
8880602 - Marking and Sealing for Unit Packages
9251374 - Carton, Packing, Ammunition for Antenna Box Assembly
ET-8851572 - Equipment Tabulation

(Copies of specifications, standards, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The antenna box assembly shall comply with all requirements specified on Drawing (dwg.) 8851572, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 Circuit resistance. The circuit resistance shall comply with table I when tested as specified in 4.4.1.

Table I
Circuit resistance (see dwg 8851572)

Receptacle	Feed thru insulator	Ohms max
Pin	Bottom insulator (section A-A)	0.1
Receptacle shell	Chassis	0.1

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3.4 Insulation resistance. The insulation resistance shall not be less than 50 megohms when subjected to a potential of 500 ± 50 volts direct current (vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.5.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-56	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Form var wire (see dwg 8851586); variable capacitor (see dwg 8851747); shielding tinned copper (see dwg 8851773); feed thru insulator (see dwg 8851580); receptacle (see dwg 8851578); varnish, moisture and fungus resistant (see MIL-V-173); screw (see AN505-2-5 and AN505-2-6, AN507-632-4, AN507-440-12); washer (see AN590C2); screw (see MS35233-30); terminal, lug (see 35430-1, 4, and 10); nut (see 35649-22 and 62); solder (see QQ-S-571); covering details of dwgs 8851572 and 8851584 as applicable.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for parts in accordance with applicable drawing or standard	Visual
Minor:		
201.	Workmanship (see 3.5)	Visual-Manual

4.3.2.1.2 Coil block (see dwg 8851585) covering a detail of dwg 8851584.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Total length, min	Gage
102.	Diameter of 2 holes, max	Gage
103.	Pitch of helix	Gage
104.	Location of 2 holes	Gage

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105. Diameter between crests of helexes.....Gage
106. Outside diameter.....Gage
107. Pitch diameter of threaded hole.....Gage
108. Minor diameter of threaded hole.....Gage
109. Material (see 3.1).....Visual

Minor:
201. Workmanship (see 3.5).....Visual-Manual

4.3.2.1.3 Assembly, coil (see dwg 8851584) covering a detail of dwg 8851572.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Length of leads, min.....*Measure
102. Wire tight on form.....Manual
*Measure with suitable scale
Minor: None defined.

4.3.2.1.4 Cover (see dwg 8851574) covering a detail of dwg 8851573.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Total height.....*Measure
102. Width of formed section.....*Measure
103. Total length.....*Measure
104. Length of flanges.....*Measure
105. Location of holes in flange.....*Measure
106. Distance from flange to end of cover (4 corners).....*Measure
107. Total width.....*Measure
108. Diameter of 4 large holes in flange.....Gage
109. Diameter of 4 small holes in flange.....Gage
110. Material (see 3.1).....Visual
*Measure with suitable scale
Minor:
201. Workmanship (see 3.5).....Visual-Manual

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4.3.2.1.5 Plate, connector (see dwg 8851577) covering a detail of dwg 8851573.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.5)	Visual-Manual

4.3.2.1.6 Cover assembly (see dwg 8851573) covering a detail of dwg 8851572.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Proper spot welds	Visual
Minor: None defined.		

4.3.2.1.7 Spacer (see dwg 8851744) covering a detail of dwg 8851572.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Length max	Gage
Minor:		
201.	Workmanship (see 3.5)	Visual

4.3.2.1.8 Assembly antenna box (see dwg 8851572).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	No parts missing or improperly assembled	Visual
102.	Soldering in accordance with MIL-S-6872	Visual-Manual
103.	Solder joints treated in accordance with MIL-T-152	Visual
104.	Proper wiring	Visual

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Minor:
201. Workmanship (see 3.5).....Visual-Manual

4.3.2.1.9 Preparation for delivery (see section 5.1).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Packaging, packing and marking in accordance with MIL-STD-1169(MU).....Visual

Minor:
201. Workmanship (see 3.5).....Visual

4.3.2.1.10 Carton, packing, ammunition, for antenna box assembly (see dwg 9251374 and section 5.2).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Item correctly wrapped and sealed with tape.....Visual
102. Bag sealed in accordance with 8880602.....Visual-Manual
103. Bag packed tight in carton.....Manual
104. Carton sealed in accordance with 8880602.....Visual-Manual
105. Marking.....Visual
106. Packaging material (see 3.1).....Visual

Minor: None defined.

4.3.3 Testing. All the tests in this paragraph shall be performed.

	Test	Test Classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance (see 3.3) and insulation resistance (see 3.4). Each assembly shall be tested as specified in 4.4.1 and 4.4.2 as applicable. If any assembly fails to comply with 3.3 and 3.4, the assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

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4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8851572. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test Methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.3). Each assembly shall be tested in accordance with table I with approved circuit resistance measuring equipment.

4.4.2 Insulation resistance (see 3.4). Each assembly shall be tested in accordance with table IV with approved insulation resistance measuring equipment.

Table IV
Insulation resistance (see dwg 8851572)

Positive Potential	Negative potential to all other pins, shell and insulator except
Pin	Bottom Insulator (section A-A)
Receptacle shell	Chassis

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

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5.1.2 Packing.

5.1.2.1 Levels A and C. The assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The assembly shall be packaged in accordance with dwg 9251374 using the specified moisture barrier.

5.2.1.2 Level C. The assembly shall be packaged in accordance with dwg 9251374 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The assembly shall be packed in accordance with dwg 9251374.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 9251374.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Production or repair part.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project No. 1195-A055

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-50297(MU) Assembly Antenna Box

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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